This `profit` function won't work yet because `revenue` and `cost` aren't defined. The next thing to do, then, is define them:

```
revenue = (salePrice) ->
  (numberSold salePrice) * salePrice

cost = (salePrice) ->
  overhead + (numberSold salePrice) * costPrice
```

These `revenue` and `cost` functions won't work until `overhead`, `numberSold`, and `costPrice` are defined. Define them:

```
overhead = 140
costPrice = 100

numberSold = (salePrice) ->
  50 + 20/10 * (200 - salePrice)
```

The formula that Agtron worked out for the number of items sold at a given sale price

There's nothing left to define. You have a solution:

```
overhead = 140
costPrice = 100

numberSold = (salePrice) ->
  50 + 20/10 * (200 - salePrice)

revenue = (salePrice) ->
  (numberSold salePrice) * salePrice

cost = (salePrice) ->
  overhead + (numberSold salePrice) * costPrice

profit = (salePrice) ->
  (revenue salePrice) – (cost salePrice)
```

Functions can be defined anywhere, so `numberSold`, `revenue`, and `cost` could be defined inside the `profit` function. When you do this, as shown in the following listing, they're scoped inside the `profit` function, meaning that they're encapsulated.

Listing 6.1. Profit from selling PhotomakerExtreme